Genesis OS / Relationship Scores

Cluster Rule Pack Specification v1

Developer Contract for Contradiction Clusters, Reflection Clusters, Severity Escalation, Write-Back Logic, and Forecast / Readiness Integration

Purpose

This document standardizes how clusters are identified, scored, escalated, written back to platform state, surfaced to AI services, and suppressed from end-user views where confidence or safety do not justify exposure. It is intended to remove implementation ambiguity across assessment services, chat telemetry, Sanctuary flows, forecast engines, and admin tooling.

Status: Build-Governing Technical Specification

Applies to: Readiness, Forecast, Drift, Shadow Activation, Sanctuary, Knock, Messaging, and AI Insight Services

1. Scope and Governing Intent

This specification defines the canonical rule-pack framework for all cluster-based detection inside Genesis OS and Relationship Scores. A cluster is a named, governed pattern created when multiple inputs suggest tension, contradiction, burden, behavioral drift, or relational instability that should influence platform logic.

This document is binding on assessment scoring services, read-model builders, forecast services, AI insight generators, Sanctuary reflection flows, messaging interventions, and admin review tooling.

Implementation Principle

Clusters are not freeform interpretations. They are governed rule objects with explicit inputs, severity rules, persistence behavior, and write-back outputs. No service may create ad hoc clusters outside the catalog without versioned approval.

2. Definitions

Term	Meaning
Cluster	A governed pattern object triggered by defined combinations of assessment answers, telemetry, behavioral events, or companion-system outputs.
Reflection Cluster	A cluster that may surface a guided self-check, Sanctuary prompt, or cooldown ritual before the system fully clears the state.
Contradiction Cluster	A cluster created when two or more signals materially conflict, suggesting inconsistency, avoidance, false-positive security, or unstable self-reporting.
Severity	The standardized level S1-S5 assigned to cluster impact based on evidence quality, persistence, risk, and recurrence.
Write-back	The persisted state change written to user, pair, forecast, drift, or AI memory ledgers after trigger conditions are satisfied.
Decay	The controlled reduction of severity over time when no reinforcing evidence appears.
Suppression	The rule that hides a cluster from user-facing surfaces even if it is active internally.

3. Cluster Families

Each cluster belongs to one primary family. Families govern default write-back targets, user visibility posture, and expected downstream effects.

ID	Family	Primary Purpose	Typical Downstream Effects
CF-CONTRA	Contradiction	Assessment + telemetry signals in tension	Readiness confidence, reflection prompts, AI caution
CF-BURDEN	Burden / Load	Accumulating emotional strain or unresolved weight	Sanctuary prompts, moderation of optimism, forecast stability

ID	Family	Primary Purpose	Typical Downstream Effects
CF-DRIFT	Drift	Behavioral movement away from prior secure or aligned state	Drift state engine, forecast caution, message intervention
CF-REPAIR	Conflict / Repair	Rupture without repair, or repair effort mismatch	Pair forecast, coaching prompts, pair-state stability
CF-SHADOW	Shadow Activation	Escalation into shadow-type behavior patterns	Safety gating, AI restraint, stronger write-back
CF-READ	Readiness Integrity	Signals that the public readiness surface may be overstated or under-supported	Score suppression, confidence downgrade

4. Core Data Contract for Every Cluster

Every cluster in the catalog must expose the same minimum fields. Optional fields may be added by family, but the minimum contract cannot be skipped.

Field	Type	Requirement
cluster_id	string	Canonical immutable identifier, e.g. CL_VULN_LOOP
cluster_name	string	Human-readable governed name
family_id	enum	One of the approved family codes
scope	enum	individual, pair, or hybrid
source_set	json	Approved input sources and field references
trigger_version	string	Rule version that created current state
severity	enum	S1-S5 current severity
evidence_count	integer	How many qualified evidence items were used
confidence_band	enum	low, guarded, medium, high
user_visible	boolean	Whether any derived user-facing label may be shown
writeback_target	enum/list	Profile, pair state, readiness ledger, forecast ledger, AI context, etc.
opened_at	timestamp	First activation time
last_reinforced_at	timestamp	Most recent supporting event
expires_at / decay_at	timestamp	Next review or decay checkpoint
clear_condition_code	enum	How the state resolves or decays

5. Severity Ladder (Universal)

The severity ladder is platform-wide. Families may tune effects, but they may not redefine the meaning of S1-S5.

Level	Meaning	Default Effects	Visibility Posture	Persistence Posture
S1	Informational pattern	No score penalty by default; log only or	Usually internal only	Fast decay; non-persistent unless

Level	Meaning	Default Effects	Visibility Posture	Persistence Posture
		soft reflection		repeated
S2	Mild tension	Light readiness confidence drag or prompt	Softened label allowed if confidence >= medium	Writes only after minimum evidence is met
S3	Active contradiction / instability	Moderate penalty or pair caution; may trigger cooldown	User may see softened insight; partner never sees raw label	Persistent until decay or reflection completion
S4	Repeating destabilizer	Strong readiness drag, forecast caution, AI restraint	Mostly internal; derived user prompt allowed	Persistent state with audit trail and recurrence memory
S5	Critical pattern	Suppression, gating, safety routing, high forecast impact	Internal only except safe intervention copy	Requires manual-safe clearing path or strong decay logic

Severity Escalation Rule

Escalation is not only about score weight. A cluster may rise in severity because of repetition, recency, cross-source confirmation, or movement from self-report conflict into observed behavior conflict.

6. Evidence Model and Trigger Standards

No cluster may activate from a single weak hint unless explicitly marked as an exception in the catalog. Each cluster rule pack must declare source quality and minimum evidence standards.

Tier	Evidence Quality	Typical Composition	Default Confidence Ceiling
Q1	Assessment only	Question-answer combinations, contradiction pairs, weighted answer families	Low to guarded
Q2	Assessment + one telemetry source	Assessment contradiction reinforced by chat retreat, ghosting, avoidance, or burden event	Guarded to medium
Q3	Two live behavioral sources	Messaging + Sanctuary or messaging + drift	Medium
Q4	Cross-family convergence	Assessment + telemetry + companion system	High
Q5	Manual review confirmed	Admin or clinician-style reviewer confirms the pattern according to policy	High

Minimum trigger requirements should be declared per cluster. As a default: S1 may open at Q1, S2 normally requires Q1 or Q2, S3 should require at least Q2, S4 should require Q3 or strong repetition, and S5 should require Q4 or an approved exception path.

7. Write-Back Policy

Write-back is the persisted operational consequence of a cluster. Triggering and write-back are related but not identical. A cluster may briefly activate in memory without writing back if evidence is insufficient.

Write-Back Class	When Used	Allowed Targets	Typical Persistence
Ephemeral observation	S1; single low-tier event	No durable write-back; event log only	24-72 hours
Soft write-back	S2 with minimum evidence	Readiness confidence buffer, reflection queue, AI caution context	7-21 days or until decay
Operational write-back	S3 or repeated S2	User cluster ledger, pair-state caution, drift interplay, forecast notes	14-45 days unless cleared
Persistent write-back	S4-S5	Primary state ledgers, gating systems, admin audit trail, suppression flags	Versioned until explicit decay / repair / safe-clear

Every write-back must include: cluster_id, severity, source snapshot hash or event references, confidence band, trigger_version, opened_at, last_reinforced_at, and the exact services allowed to consume the record.

8. Decay, Clearing, and Downgrade Rules

Clusters should not become immortal. Every cluster family must define decay checkpoints, downgrade conditions, and hard clear logic.

Clear/Downgrade Path	Condition	Result
Time decay	No reinforcing events occur within the review window	Severity may drop by one level only if no hard block exists
Reflection completion	User completes required reflection and no contradictory reinforcement appears	May reduce visibility or severity; does not always fully clear
Repair evidence	Behavioral evidence shows improvement or successful repair	Pair and drift clusters may downgrade more quickly
Contradiction resolution	New assessment or pattern evidence removes the original conflict	Cluster may clear if confidence in resolution >= medium
Manual safe-clear	Admin-approved correction of false positive or policy exception	Full clear with audit note; never silent delete

9. Visibility and Exposure Policy

Raw cluster names are not a consumer feature. User-facing exposure should use softened, safe-language translations unless a product surface explicitly requires more direct naming and policy approves it.

Surface	What It May Receive	Allowed?
Internal engine services	Raw cluster_id + severity + evidence metadata	Yes
Admin tooling	Raw cluster objects + audit + rule	Yes

Surface	What It May Receive	Allowed?
	version	
AI insight services	Derived interpretation with direct-name permission flags	Conditional
User readiness surfaces	Softened themes only, never raw contradiction law	Conditional
Match partner surfaces	No direct cluster exposure; only safe derived caution states where policy permits	Highly restricted

Non-Negotiable Exposure Rule

A partner must never see another user's raw cluster catalog, severity stack, contradiction labels, or shadow activation state. Pair-facing outputs must be translated into safe, minimal-risk guidance.

10. Readiness and Forecast Interaction Map

Clusters influence readiness and forecast in different ways. Some reduce score. Some reduce confidence only. Some suppress public display without changing the underlying internal model.

Family	May Change Readiness	May Change Forecast	May Trigger Suppression	May Trigger Intervention
CF-CONTRA	Yes	Yes	Sometimes	Yes
CF-BURDEN	Yes	Yes	Sometimes	Conditional
CF-DRIFT	Indirect to direct	Yes	Yes	Yes
CF-REPAIR	Rarely direct	Yes	No	Conditional
CF-SHADOW	Yes	Yes	Yes	Yes
CF-READ	Yes	No or limited	Yes	Yes

11. Cluster Catalog v1 (Initial Governed Set)

The following catalog establishes the initial governed clusters for v1 implementation. Additional clusters require a versioned addendum, migration notes, and consumer-service impact review.

ID	Name	Family	Scope	Core Trigger Theme	Range
CL_VULN_LOOP	Vulnerability Loop	CF-CONTRA	Individual	Q6 + Q32 contradictions; optional retreat telemetry	S2-S4
CL_CONTROL_AVOID	Control Avoidance	CF-CONTRA	Individual	Claims openness while preserving high control / low surrender safety	S2-S4
CL_PUSH_PULL	Push-Pull Attachment Cycle	CF-DRIFT	Hybrid	Approach bursts followed by withdrawal, ghosting, or inconsistent message	S2-S5

ID	Name	Family	Scope	Core Trigger Theme	Range
				rhythm	
CL_REPAIR_GAP	Repair Gap	CF-REPAIR	Pair	Conflict occurs without repair attempt or with mismatch in repair effort	S2-S4
CL_BURDEN_STACK	Burden Stack	CF-BURDEN	Individual	Accumulation across grief, anger, fatigue, overload, or unresolved emotional labor	S1-S4
CL_SHADOW_SURGE	Shadow Surge	CF-SHADOW	Hybrid	Short-window pattern spike matching shadow rules	S3-S5
CL_READY_OVR	Readiness Overstatement	CF-READ	Individual	Public-facing readiness unsupported by recent evidence or contradicted by active clusters	S2-S4
CL_SIGNAL_SPARSITY	Signal Sparsity	CF-READ	Individual/Pair	Too little evidence to justify stable public conclusions	S1-S3
CL_DRIFT_SLOW	Slow Drift	CF-DRIFT	Hybrid	Gradual decline in reciprocity, rhythm, or response quality over windowed periods	S2-S4
CL_FORECAST_FOG	Forecast Fog	CF-READ	Pair	Forecast confidence degraded by conflicting or stale inputs	S1-S3

12. Rule Pack Template (Required for Every Cluster)

Each cluster must have its own subordinate rule pack. The template below is mandatory and should be stored in code-adjacent configuration or a governed rules repository.

Rule Pack Section	What Must Be Defined
Identity block	cluster_id, cluster_name, family, scope, version, owner service
Input block	source fields, acceptable event types, lookback windows, exclusions

Rule Pack Section	What Must Be Defined
Trigger block	minimum evidence, event combinations, threshold math, severity open rules
Escalation block	how repetition, recency, and cross-source confirmation raise severity
Write-back block	targets, payload shape, persistence class, idempotency behavior
Decay block	time-based downgrade windows, reflection effects, repair effects
Visibility block	what surfaces can see raw or softened output
Score / forecast block	what families may be influenced and whether effect is direct, indirect, or suppressive
Test block	positive cases, negative cases, border cases, false positive guards

13. Standard Threshold Guidance

The exact score math may live in a workbook or service-specific appendix, but the operational thresholds below should govern default cluster behavior unless a cluster-specific exception is approved.

Threshold Type	Default Guidance
Open threshold	At least 1 qualifying rule path satisfied
Write-back threshold	At least medium confidence or approved S5 exception
Escalation threshold	Reinforcement within review window or cross-source convergence
Suppression threshold	S3+ when public surface would overstate certainty or safety
Manual review threshold	S5 or policy-marked clusters with consumer impact

14. AI Consumption Rules

AI services may consume cluster outputs for coaching, reflection writing, prompt shaping, or intervention timing. AI may not improvise new cluster labels, intensify severity, or expose raw cluster law to users without permission flags.

AI Rule Class	Requirement
Allowed	Use softened cluster themes to tailor reflections, caution optimistic forecast phrasing, request more information, or recommend Sanctuary flows.
Restricted	Mention raw cluster names only in internal admin or developer contexts.
Forbidden	State or imply diagnosis, certainty of future failure, or invisible trait claims unsupported by the evidence contract.

15. Drift Interlock Requirements

Because drift is one of the most consequential families for readiness and forecast, all drift clusters must interlock with DriftEligibility logic rather than bypassing it.

Required drift interlocks: (a) drift clusters must record the rollup window used, (b) forecast services must know whether the drift state is eligible, emerging, active, or suppressed, (c) readiness surfaces must not overreact to early drift without confidence support, and (d) shadow activation may intensify drift severity only when the shadow rules agree.

Default Guardrail

Slow drift may be tracked internally before it is made visible. Visibility should lag internal detection until the platform has sufficient recurrence, cross-source support, or repair failure evidence.

16. Testing and QA Requirements

QA Layer	Requirement
Unit tests	Every cluster rule pack needs deterministic trigger and non-trigger cases
Regression tests	Changing a threshold must show before/after impact on historical samples
False positive tests	Rule packs must include examples that should not open despite superficial similarity
Cross-service parity tests	Read model, forecast service, and AI context builder must agree on active cluster state
Audit sampling	Admin review should inspect a sampled percentage of S4-S5 activations

17. Implementation Sequence

- Freeze the cluster catalog and assign immutable IDs.
- Implement the universal severity ladder and write-back classes.
- Ship rule packs first for: Vulnerability Loop, Control Avoidance, Slow Drift, Push-Pull, Repair Gap, and Readiness Overstatement.
- Attach softened user-facing language and suppression rules before exposing any cluster-derived surfaces publicly.
- Run historical simulation and false-positive review before enabling forecast penalties in production.
- Version every future cluster expansion with migration notes and parity tests.

18. Required Appendices After v1

This specification is implementation-ready for governance, but the following appendices should be maintained beside it as the system matures: (1) cluster-by-cluster rule pack files, (2) threshold workbook references, (3) user-facing copy libraries, (4) admin review playbooks, and (5) historical migration notes for rule changes.